Supplementary Proofs for VERMITHOR:

Formally Verified Runtime Orchestration
for Thermally Safe Edge CPS Inference

This document provides the complete proofs for all theorems, propositions, and formal results stated in the main paper. The numbering and notation follow the main text exactly. Throughout, we use the definitions, assumptions, and notation established in Sections III and IV of the main paper.

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1 Preliminaries: Definitions and Assumptions

For self-containedness, we restate the key definitions and assumptions from the main paper that are used across the proofs below.

Definition 1 (Hybrid State)**.** The composite system state at discrete control step $t \in \mathbb{N}$ is $\mathbf{x}_t = (T_t, \mathbf{x}_t^{\text{net}}, m_t)$, where $T_t \in \mathbb{R}_{\geq 0}$ is the GPU junction temperature, $\mathbf{x}_t^{\text{net}} \in \mathbb{R}^d$ captures the network sub-state (e.g., estimated RTT, packet loss rate, jitter), and $m_t \in \mathcal{M} = \{m_1, \ldots, m_5\}$ is the active inference mode.

Definition 2 (Signal Trajectory and Evaluation Window)**.** At each step t, a past-time evaluation window $TW(t, W) = \{t - W, \ldots, t-1, t\} \subset \mathbb{N}$ specifies a historical horizon of W steps. The signal trajectory $\sigma = (\mathbf{x}_0, \mathbf{x}_1, \ldots)$ records the full evolution of the hybrid state, and the past-time STL (ptSTL) specifications are evaluated over $TW(t, W)$.

Definition 3 (STL Robustness Degree)**.** Given a ptSTL specification φ, a signal trajectory σ, and a step t, the robustness degree $\rho(\varphi, \sigma, t) \in \mathbb{R}$ quantifies the signed margin of satisfaction: $\rho > 0$ indicates strict satisfaction, $\rho = 0$ is the boundary, and $\rho < 0$ constitutes a formal violation.

Assumption 4 (Bounded Thermal Rate)**.** Due to hardware-specific heat capacity, the derivative of the GPU temperature is bounded by a known Lipschitz constant $L_{\max} > 0$:

$$|\dot{T}_t| \leq L_{\max}. \tag{1}$$

Empirically, $L_{\max} = 0.083\,^{\circ}\mathrm{C/s}$ on the NVIDIA Jetson Thor (Section V-A of main paper).

Definition 5 (f-Divergence Inflated Prediction Region)**.** Let $D_f(\mathcal{P}_{\text{run}} \| \mathcal{P}_{\text{cal}})$ denote the divergence between the runtime and calibration distributions, instantiated as the KL divergence D_{KL} in our framework. The Robust Prediction Region $\Gamma_t^{\varepsilon, D_f}$ dynamically inflates the nominal CP interval $[q - w, q + w]$ by a factor $\lambda(D_f)$:

$$\Gamma_t^{\varepsilon, D_f} = \big[q - \lambda(D_f) \cdot w,\ q + \lambda(D_f) \cdot w\big], \tag{2}$$

where q is the conformal quantile from calibration, w is the base conformal half-width, $\lambda(D_f) = \min(\lambda_{\max},\, 1 + \beta \cdot D_f)$, $\beta > 0$ is the inflation sensitivity, and $\lambda_{\max}$ is the maximum permissible inflation factor.

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2 Proof of Theorem 1: Lipschitz Physical Robustness

Theorem 6 (Lipschitz Physical Robustness — restated)**.** *Under Assumption 4, the thermal safety predicate $\mu_{\text{ph}}(\mathbf{x}_t) = T_{\text{lim}} - T_t$ is Lipschitz continuous with constant $L_{\max}$:*

$$|\mu_{\text{ph}}(\mathbf{x}_{t'}) - \mu_{\text{ph}}(\mathbf{x}_t)| \leq L_{\max}\, |t' - t|. \tag{3}$$

Consequently, if $\rho(\varphi_{\text{ph}}, \sigma, t) > 0$, the system is guaranteed to remain thermally safe for a strictly bounded predictive horizon:

$$\Delta t^{\text{safe}} \geq \frac{\rho(\varphi_{\text{ph}}, \sigma, t)}{L_{\max}}. \tag{4}$$

Proof. We proceed in two parts.

Part 1: Lipschitz continuity of μ_{ph}. The thermal safety predicate is $\mu_{\text{ph}}(\mathbf{x}_t) = T_{\text{lim}} - T_t$, where T_{lim} is a constant (the thermal throttling threshold). Therefore:

$$|\mu_{\text{ph}}(\mathbf{x}_{t'}) - \mu_{\text{ph}}(\mathbf{x}_t)| = |(T_{\text{lim}} - T_{t'}) - (T_{\text{lim}} - T_t)| = |T_t - T_{t'}|. \tag{5}$$

By Assumption 4, $|\dot{T}_t| \leq L_{\max}$. For any two time steps $t, t' \in \mathbb{N}$ with $t' > t$, integrating the bounded rate of change:

$$|T_{t'} - T_t| = \left| \int_t^{t'} \dot{T}_s\, ds \right| \leq \int_t^{t'} |\dot{T}_s|\, ds \leq L_{\max} \cdot (t' - t) = L_{\max}\, |t' - t|. \tag{6}$$

Combining (5) and (6):

$$|\mu_{\text{ph}}(\mathbf{x}_{t'}) - \mu_{\text{ph}}(\mathbf{x}_t)| \leq L_{\max}\, |t' - t|, \tag{7}$$

which establishes (3).

Part 2: Predictive safety horizon. The thermal safety specification is $\varphi_{\text{ph}} := \boxminus_{[0,W]}(\mu_{\text{ph}}(\mathbf{x}_t) \geq 0)$. The robustness degree of φ_{ph} at the current step is:

$$\rho(\varphi_{\text{ph}}, \sigma, t) = \min_{k \in [0,W]} \mu_{\text{ph}}(\mathbf{x}_{t-k}). \tag{8}$$

Suppose $\rho(\varphi_{\text{ph}}, \sigma, t) > 0$. This implies $\mu_{\text{ph}}(\mathbf{x}_t) \geq \rho(\varphi_{\text{ph}}, \sigma, t) > 0$, and in particular $\mu_{\text{ph}}(\mathbf{x}_t) > 0$ (the system is currently thermally safe).

We seek the earliest future time $t' > t$ at which $\mu_{\text{ph}}(\mathbf{x}_{t'})$ could reach zero. By Part 1, the predicate value can decrease at most at rate $L_{\max}$ per unit time. Setting the worst-case trajectory $\mu_{\text{ph}}(\mathbf{x}_{t'}) = \mu_{\text{ph}}(\mathbf{x}_t) - L_{\max}(t' - t)$ and solving for the zero-crossing:

$$\mu_{\text{ph}}(\mathbf{x}_t) - L_{\max} \cdot \Delta t^{\text{safe}} = 0 \tag{9}$$

$$\Delta t^{\text{safe}} = \frac{\mu_{\text{ph}}(\mathbf{x}_t)}{L_{\max}} \geq \frac{\rho(\varphi_{\text{ph}}, \sigma, t)}{L_{\max}}, \tag{10}$$

where the inequality follows because $\mu_{\text{ph}}(\mathbf{x}_t) \geq \rho(\varphi_{\text{ph}}, \sigma, t)$ by (8). This establishes (4). □

Remark 7. *The safe horizon provides a forward-looking guarantee from historical data: the controller can predict how long thermal safety is guaranteed to persist under worst-case heating, enabling proactive mode switching before any violation occurs. The bound is tight in the sense that there exist thermal trajectories (monotonic heating at rate $L_{\max}$) that achieve equality.*

3 Justification of Proposition 1: Stochastic Network Dynamics

Proposition 8 (Stochastic Network Dynamics — restated)**.** *The network sub-state $\mathbf{x}_t^{\text{net}}$ undergoes abrupt, non-Lipschitz transitions (e.g., tunnel entry, sudden interference). Therefore, $\rho_{\text{net}}(t)$ cannot be bounded deterministically. Instead, it must be evaluated via worst-case projection over the Robust Prediction Region $\Gamma_t^{\varepsilon, D_f}$ (Definition 5):*

$$\tilde{\rho}_{\text{net}}(t) = \inf_{\hat{\mathbf{x}}^{\text{net}} \in \Gamma_t^{\varepsilon, D_f}} \rho(\varphi_{\text{net}}, \hat{\mathbf{x}}^{\text{net}}, t). \tag{11}$$

Justification. This proposition is a construction that follows from the fundamental distinction between the physical and network subsystems.

Non-Lipschitz nature. Unlike the thermal state, which is constrained by the heat equation and thus satisfies the Lipschitz bound of Assumption 4, the network state $\mathbf{x}_t^{\text{net}}$ (RTT, packet loss, jitter) can undergo discontinuous transitions. For example, a mobile device entering a tunnel experiences an instantaneous jump from nominal to degraded network conditions. Formally, for any constant $L > 0$, there exist consecutive time steps $t, t+1$ such that:

$$\|\mathbf{x}_{t+1}^{\text{net}} - \mathbf{x}_t^{\text{net}}\| > L, \tag{12}$$

precluding any Lipschitz-type bound.

Necessity of worst-case projection. Since no deterministic bound on $\mathbf{x}_t^{\text{net}}$ is available, the controller cannot guarantee $\rho_{\text{net}}(t) \geq 0$ with certainty for the *true* realization. Instead, we leverage the Robust Prediction Region $\Gamma_t^{\varepsilon, D_f}$, which contains the true state with high probability (quantified in Theorem 11). By evaluating robustness over the *worst case* within this region, i.e., taking the infimum in (11), we ensure that:

- If $\tilde{\rho}_{\text{net}}(t) \geq 0$, then $\rho(\varphi_{\text{net}}, \mathbf{x}_t^{\text{net}}, t) \geq 0$ for *every* possible realization within $\Gamma_t^{\varepsilon, D_f}$.
- The probability that $\mathbf{x}_t^{\text{net}} \in \Gamma_t^{\varepsilon, D_f}$ is at least $1 - \varepsilon - \delta(D^*)$ by Theorem 11.

This converts the stochastic network uncertainty into a probabilistic safety guarantee, bridging the gap between the deterministic thermal subsystem and the stochastic network subsystem.

Computational realization. For the one-dimensional network predicates used in VERMITHOR (e.g., $\mu_{\text{net}}(\mathbf{x}_t^{\text{net}}) = \mathbf{x}_t^{\text{net}} - \ell_{\min}$ for latency lower bound), the infimum over a symmetric interval $\Gamma_t^{\varepsilon, D_f} = [q - \lambda w, q + \lambda w]$ is achieved at the boundary:

$$\tilde{\rho}_{\text{net}}(t) = \mu_{\text{net}}(q - \lambda w) = (q - \lambda w) - \ell_{\min}, \tag{13}$$

requiring $O(1)$ computation per predicate. □

4 Proof of Theorem 2: Hybrid Robustness by Conjunction

Theorem 9 (Hybrid Robustness by Conjunction — restated)**.** *By the quantitative ptSTL conjunction semantics, the composite specification $\varphi_{\text{ph}} \wedge \varphi_{\text{net}}$ yields the hybrid robustness:*

$$\rho_{\text{hybrid}}(t) = \min\big(\rho_{\text{ph}}(t),\ \tilde{\rho}_{\text{net}}(t)\big). \tag{14}$$

A controller maintaining $\rho_{\text{hybrid}}(t) \geq 0$ guarantees joint satisfaction of the deterministic physical limits and the probabilistic network bounds at step t.

Proof. The proof follows from the standard quantitative semantics of Signal Temporal Logic [1, 3].

Step 1: STL conjunction semantics. The quantitative semantics of STL define the robustness of a conjunction as:

$$\rho(\varphi_1 \wedge \varphi_2, \sigma, t) = \min\big(\rho(\varphi_1, \sigma, t),\ \rho(\varphi_2, \sigma, t)\big). \tag{15}$$

This is a foundational property of quantitative STL semantics, established in [1] and [2]. The intuition is that a conjunction is satisfied if and only if both conjuncts are satisfied; the quantitative margin of satisfaction is determined by the "weaker" conjunct.

Step 2: Application to the hybrid specification. Let φ_{ph} denote the thermal safety specification with robustness $\rho_{\text{ph}}(t) = \rho(\varphi_{\text{ph}}, \sigma, t)$, and let φ_{net} denote the network specification with worst-case projected robustness $\tilde{\rho}_{\text{net}}(t) = \rho(\varphi_{\text{net}}, \hat{\sigma}, t)$ evaluated over $\Gamma_t^{\varepsilon, D_f}$ (Proposition 8).

Applying (15) to the composite specification $\varphi_{\text{safe}} := \varphi_{\text{ph}} \wedge \varphi_{\text{net}}$:

$$\rho_{\text{hybrid}}(t) := \rho(\varphi_{\text{safe}}, \sigma, t) = \min\big(\rho_{\text{ph}}(t),\ \tilde{\rho}_{\text{net}}(t)\big). \tag{16}$$

Step 3: Joint satisfaction guarantee. If the controller maintains $\rho_{\text{hybrid}}(t) \geq 0$, then:

$$\min\big(\rho_{\text{ph}}(t),\ \tilde{\rho}_{\text{net}}(t)\big) \geq 0 \tag{17}$$

$$\Longrightarrow \quad \rho_{\text{ph}}(t) \geq 0 \quad \text{and} \quad \tilde{\rho}_{\text{net}}(t) \geq 0. \tag{18}$$

The first condition guarantees thermal safety (deterministically, by Theorem 6). The second condition guarantees that network robustness holds for all realizations within $\Gamma_t^{\varepsilon, D_f}$ (by construction of the worst-case projection in Proposition 8). Combined with the coverage guarantee of Theorem 11, this establishes joint satisfaction. □

Remark 10 (Extension to Three-Term Conjunction)**.** *The main paper's orchestration algorithm (Algorithm 2) evaluates the full composite specification* $\varphi_{\text{safe}} = \varphi_{\text{ph}} \wedge \varphi_{\text{net}} \wedge \varphi_{\text{dwell}}$*, yielding:*

$$\rho(\varphi_{\text{safe}}, \sigma, t) = \min\big(\rho_{\text{ph}}(t),\ \tilde{\rho}_{\text{net}}(t),\ \rho_{\text{dwell}}(t)\big). \tag{19}$$

This follows from the same conjunction semantics applied to three conjuncts: $\min(a, b, c) = \min(\min(a, b), c)$*. Since* φ_{dwell} *is enforced deterministically by the controller logic (the dwell-time counter prevents mode switching within* $\tau_{\min}$ *steps), the probabilistic bound remains governed solely by the network coverage term.*

5 Proof of Theorem 3: Coverage under Distribution Shift

Theorem 11 (Coverage under Distribution Shift — restated)**.** *Given a bounded KL divergence* $D_{\text{KL}}(\mathcal{P}_{\text{run}} \| \mathcal{P}_{\text{cal}}) \leq D^*$*, the robust interval satisfies the marginal coverage target with an analytically bounded miscoverage penalty:*

$$\Pr\big[\mathbf{x}_t^{\text{net}} \in \Gamma_t^{\varepsilon, D_f}\big] \geq 1 - \varepsilon - \delta(D^*), \tag{20}$$

where $\delta(D^*) = \sqrt{D^*/2}$ *follows from Pinsker's inequality.*

Proof. The proof proceeds in three steps: (1) we recall the standard conformal prediction guarantee, (2) we bound the coverage degradation under distribution shift via Pinsker's inequality, and (3) we show that the adaptive inflation recovers the target coverage.

Step 1: Standard conformal prediction guarantee. Under the exchangeability assumption (i.e., when $\mathcal{P}_{\text{run}} = \mathcal{P}_{\text{cal}}$), standard conformal prediction [5] constructs a prediction set Γ_t^{ε} satisfying:

$$\Pr_{\mathcal{P}_{\text{cal}}}\big[\mathbf{x}_t^{\text{net}} \in \Gamma_t^{\varepsilon}\big] \geq 1 - \varepsilon. \tag{21}$$

This guarantee is distribution-free and holds for any marginal distribution $\mathcal{P}_{\text{cal}}$, relying only on the exchangeability of the calibration and test data.

Step 2: Coverage degradation under shift. When the runtime distribution $\mathcal{P}_{\text{run}}$ differs from the calibration distribution $\mathcal{P}_{\text{cal}}$, the coverage guarantee (21) no longer holds. We bound the degradation as follows.

Let $A = \Gamma_t^{\varepsilon}$ denote the nominal conformal prediction set. The coverage under the shifted distribution is:

$$\Pr_{\mathcal{P}_{\text{run}}}\big[\mathbf{x}_t^{\text{net}} \in A\big] = \Pr_{\mathcal{P}_{\text{cal}}}\big[\mathbf{x}_t^{\text{net}} \in A\big] + \big(\Pr_{\mathcal{P}_{\text{run}}}[A] - \Pr_{\mathcal{P}_{\text{cal}}}[A]\big). \tag{22}$$

By the definition of total variation distance:

$$\text{TV}(\mathcal{P}_{\text{run}}, \mathcal{P}_{\text{cal}}) := \sup_{A \in \mathcal{F}} \big| \Pr_{\mathcal{P}_{\text{run}}}[A] - \Pr_{\mathcal{P}_{\text{cal}}}[A] \big|, \tag{23}$$

the coverage difference for any event A is bounded:

$$\Pr_{\mathcal{P}_{\text{run}}}[A] - \Pr_{\mathcal{P}_{\text{cal}}}[A] \geq -\text{TV}(\mathcal{P}_{\text{run}}, \mathcal{P}_{\text{cal}}). \tag{24}$$

Pinsker's inequality [4] relates the total variation distance to the KL divergence:

$$\text{TV}(\mathcal{P}_{\text{run}}, \mathcal{P}_{\text{cal}}) \leq \sqrt{\frac{D_{\text{KL}}(\mathcal{P}_{\text{run}} \| \mathcal{P}_{\text{cal}})}{2}}. \tag{25}$$

Under the assumption $D_{\mathrm{KL}}(\mathcal{P}_{\mathrm{run}} \| \mathcal{P}_{\mathrm{cal}}) \leq D^*$, we obtain:

$$\mathrm{TV}(\mathcal{P}_{\mathrm{run}}, \mathcal{P}_{\mathrm{cal}}) \leq \sqrt{\frac{D^*}{2}} =: \delta(D^*). \tag{26}$$

Combining (21), (22), and (24):

$$\begin{aligned} \Pr_{\mathcal{P}_{\mathrm{run}}}\left[\mathbf{x}_t^{\mathrm{net}} \in \Gamma_t^{\varepsilon}\right] &\geq \Pr_{\mathcal{P}_{\mathrm{cal}}}\left[\mathbf{x}_t^{\mathrm{net}} \in \Gamma_t^{\varepsilon}\right] - \mathrm{TV}(\mathcal{P}_{\mathrm{run}}, \mathcal{P}_{\mathrm{cal}}) \\ &\geq (1-\varepsilon) - \delta(D^*) \\ &= 1 - \varepsilon - \delta(D^*). \end{aligned} \tag{27}$$

This establishes the coverage bound for the *nominal* (uninflated) prediction set.

Step 3: Role of adaptive inflation. The bound (27) shows that the nominal set Γ_t^{ε} loses coverage by at most $\delta(D^*)$ under shift. The inflated set $\Gamma_t^{\varepsilon, D_f} \supseteq \Gamma_t^{\varepsilon}$ (since $\lambda(D_f) \geq 1$ whenever $D_f > 0$) can only *increase* coverage:

$$\Pr_{\mathcal{P}_{\mathrm{run}}}\left[\mathbf{x}_t^{\mathrm{net}} \in \Gamma_t^{\varepsilon, D_f}\right] \geq \Pr_{\mathcal{P}_{\mathrm{run}}}\left[\mathbf{x}_t^{\mathrm{net}} \in \Gamma_t^{\varepsilon}\right] \geq 1 - \varepsilon - \delta(D^*), \tag{28}$$

where the first inequality follows from $\Gamma_t^{\varepsilon} \subseteq \Gamma_t^{\varepsilon, D_f}$ and the monotonicity of probability measures.

The adaptive inflation factor $\lambda(D_f) = \min(\lambda_{\max}, 1 + \beta \cdot D_f)$ serves to *recover* coverage beyond the worst-case bound of (27). Empirically (Section V-C of the main paper), the online f-divergence estimator tracks D_{KL} in real-time and inflates the prediction interval proportionally, maintaining coverage within 0.6 pp of the target $1 - \varepsilon$ even at maximal shift. □

Remark 12 (Connection to Zhao et al.)**.** *The coverage bound* (27) *is a special case of the general framework of Zhao et al. [6], who derive coverage guarantees under f-divergence bounded shifts using the function $g(\beta) = \inf_{\alpha \geq 0}\{\alpha f^*(\beta/\alpha) : \alpha \leq \varepsilon\}$ and its inverse. For the special case of KL divergence, the Pinsker bound provides a simpler and more interpretable expression. Since our online estimator (Section IV-E of the main paper) tracks D_{KL} directly, the simplified form $\delta(D^*) = \sqrt{D^*/2}$ is both sufficient and operationally natural.*

Remark 13 (Tightness)**.** *The bound $\delta(D^*) = \sqrt{D^*/2}$ from Pinsker's inequality is known to be tight: there exist distributions $\mathcal{P}_{\mathrm{run}}, \mathcal{P}_{\mathrm{cal}}$ with $D_{\mathrm{KL}} = D^*$ for which $\mathrm{TV} = \sqrt{D^*/2}$. Tighter bounds can be obtained using the Bretagnolle–Huber inequality ($\mathrm{TV} \leq \sqrt{1 - \exp(-D_{\mathrm{KL}})}$), but this refinement is of limited practical value since the online estimator introduces its own estimation error.*

6 Proof of Theorem 4: Amortized Runtime Complexity

Theorem 14 (Amortized Runtime Complexity — restated)**.** *Algorithm 2 (the VERMITHOR orchestrator) executes in $O(1)$ amortized time per control step, independent of the evaluation window length W.*

Proof. Algorithm 2 consists of four sequential operations at each control step t. We analyze each and show that it contributes $O(1)$ amortized cost.

Operation 1: Monotonic-queue STL evaluation ($O(1)$ amortized). Algorithm 1 maintains a monotonic deque Q to compute $\rho(\boxminus_{[0,W]}\varphi, \sigma, t) = \min_{k\in[0,W]} \mu(\mathbf{x}_{t-k})$ incrementally. At each step t:

(a) The current predicate value $\mu(\mathbf{x}_t)$ is computed in $O(1)$.
(b) Dominated entries (those with robustness $\geq \mu(\mathbf{x}_t)$) are removed from the back of Q.
(c) The new entry $(t, \mu(\mathbf{x}_t))$ is appended to the back.
(d) Expired entries (those with timestamp $< t - W$) are removed from the front.
(e) The front of Q yields the window minimum.

Steps (b) and (d) may each remove multiple entries in a single step, but across the lifetime of the algorithm, each entry is inserted exactly once (step (c)) and removed exactly once (either via step (b) or (d)). By the amortized argument, the total number of insertions and deletions over N steps is at most $2N$, giving $O(1)$ amortized cost per step.

VERMITHOR runs three such evaluators (for φ_{ph}, φ_{net}, and φ_{dwell}), contributing $3 \times O(1) = O(1)$ amortized cost.

Operation 2: EMA divergence update ($O(1)$ worst-case). The exponential moving average update:

$$\hat{D}_t = (1-\alpha)\hat{D}_{t-1} + \alpha \cdot D_{\text{KL}}(\text{batch}) \tag{29}$$

consists of one multiplication, one addition, and one assignment—$O(1)$ arithmetic operations.

Operation 3: PID inflation control ($O(1)$ worst-case). The PID controller computes:

$$u_t = \beta\hat{D}_t + 0.5\beta\bar{D}_t - \kappa(\lambda_{t-1} - \lambda_{\text{base}}), \qquad \lambda_t = \text{clip}(\lambda_{\text{base}} + u_t, \lambda_{\text{base}}, \lambda_{\max}), \tag{30}$$

involving a constant number of multiplications, additions, and a clip operation—$O(1)$ arithmetic.

Operation 4: Density-ratio forward pass ($O(1)$ worst-case). The lightweight neural classifier r_θ processes a fixed-size batch of network observations. Since the batch size N is a constant (independent of W and t), and the network architecture is fixed, the forward pass requires $O(1)$ time.

Operation 5: Mode selection ($O(|\mathcal{M}|)$ worst-case). The feasible mode set $\mathcal{M}_{\text{safe}}(t) = \{m \in \mathcal{M} \mid \hat{\rho}(\varphi_i, \sigma, t \mid m) \geq 0, \forall \varphi_i\}$ iterates over $|\mathcal{M}| = 5$ modes, evaluating $|\hat{\varphi}|$ specifications per mode. Each projected robustness $\hat{\rho}$ uses the Lipschitz bound (Theorem 6) for thermal projection and the worst-case evaluation (Proposition 8) for network projection, both in $O(1)$. Since $|\mathcal{M}|$ and $|\hat{\varphi}|$ are constants, this step is $O(1)$.

Total. The per-step cost is:

$$\underbrace{O(1)}_{\text{mono queues}} + \underbrace{O(1)}_{\text{EMA}} + \underbrace{O(1)}_{\text{PID}} + \underbrace{O(1)}_{r_\theta} + \underbrace{O(|\mathcal{M}| \cdot |\hat{\varphi}|)}_{\text{mode select}} = O(1) \text{ amortized}, \tag{31}$$

independent of W. □

7 Proof of Theorem 5: Probabilistic Safety Guarantee

Theorem 15 (Probabilistic Safety Guarantee — restated)**.** *Under bounded distribution shift $D_f(\mathcal{P}_{\text{run}} \| \mathcal{P}_{\text{cal}}) \leq D^*$, Algorithm 2 guarantees:*

$$\Pr\Big[\rho(\varphi_{\text{safe}}, \sigma, t) \geq 0\Big] \geq 1 - \varepsilon - \delta(D^*), \quad \forall t, \tag{32}$$

where $\delta(D^*) = \sqrt{D^*/2}$.

Proof. The composite safety specification decomposes as:

$$\varphi_{\text{safe}} = \varphi_{\text{ph}} \wedge \varphi_{\text{net}} \wedge \varphi_{\text{dwell}}. \tag{33}$$

By the conjunction semantics (Theorem 9 and its extension in Remark 4):

$$\rho(\varphi_{\text{safe}}, \sigma, t) = \min\big(\rho_{\text{ph}}(t),\ \tilde{\rho}_{\text{net}}(t),\ \rho_{\text{dwell}}(t)\big). \tag{34}$$

We analyze each sub-specification independently.

Thermal specification φ_{ph}: deterministic satisfaction. Under Theorem 6, the thermal predicate is Lipschitz continuous. Algorithm 2 selects a mode m_t^* from the feasible set $\mathcal{M}_{\text{safe}}(t)$, which by construction satisfies $\hat{\rho}(\varphi_{\text{ph}}, \sigma, t \mid m_t^*) \geq 0$. The projected robustness uses the Lipschitz bound to predict the thermal trajectory under mode m_t^*'s power profile. Since the Lipschitz bound is exact (Assumption 4), the projection is a worst-case guarantee:

$$\Pr[\rho_{\text{ph}}(t) \geq 0] = 1. \tag{35}$$

Dwell-time specification φ_{dwell}: deterministic satisfaction. The dwell-time specification requires that no mode switch occurs within $\tau_{\min}$ steps of the previous switch. This is enforced by the controller logic directly: a mode-switch counter tracks the time since the last transition, and the controller blocks all mode changes until the counter exceeds $\tau_{\min}$. Since this is a hard-coded logical constraint (not dependent on stochastic observations):

$$\Pr[\rho_{\text{dwell}}(t) \geq 0] = 1. \tag{36}$$

Network specification φ_{net}: probabilistic satisfaction. The controller evaluates $\tilde{\rho}_{\text{net}}(t) = \inf_{\hat{\mathbf{x}}^{\text{net}} \in \Gamma_t^{\varepsilon, D_f}} \rho(\varphi_{\text{net}}, \hat{\mathbf{x}}^{\text{net}}, t)$ (Proposition 8). By construction, $\tilde{\rho}_{\text{net}}(t) \geq 0$ implies $\rho(\varphi_{\text{net}}, \mathbf{x}_t^{\text{net}}, t) \geq 0$ for all $\mathbf{x}_t^{\text{net}} \in \Gamma_t^{\varepsilon, D_f}$.

By Theorem 11, the true state lies within the prediction region with probability:

$$\Pr\big[\mathbf{x}_t^{\text{net}} \in \Gamma_t^{\varepsilon, D_f}\big] \geq 1 - \varepsilon - \delta(D^*). \tag{37}$$

Therefore, whenever $\mathbf{x}_t^{\text{net}} \in \Gamma_t^{\varepsilon, D_f}$ and the controller selects a mode with $\tilde{\rho}_{\text{net}}(t) \geq 0$:

$$\Pr[\rho(\varphi_{\text{net}}, \sigma, t) \geq 0] \geq 1 - \varepsilon - \delta(D^*). \tag{38}$$

Combining the three sub-specifications. From (34), $\rho(\varphi_{\text{safe}}, \sigma, t) \geq 0$ if and only if all three sub-specifications have non-negative robustness. Since (35) and (36) hold with probability 1, the bottleneck is the network specification:

$$\begin{aligned}\Pr\big[\rho(\varphi_{\text{safe}}, \sigma, t) \geq 0\big] &= \Pr\big[\rho_{\text{ph}}(t) \geq 0 \ \wedge \ \tilde{\rho}_{\text{net}}(t) \geq 0 \ \wedge \ \rho_{\text{dwell}}(t) \geq 0\big] \\ &= \Pr\big[\tilde{\rho}_{\text{net}}(t) \geq 0\big] \cdot \underbrace{\Pr[\rho_{\text{ph}}(t) \geq 0]}_{=1} \cdot \underbrace{\Pr[\rho_{\text{dwell}}(t) \geq 0]}_{=1} \\ &\geq 1 - \varepsilon - \delta(D^*), \end{aligned} \tag{39}$$

where the second equality uses the independence of the deterministic and probabilistic terms, and the inequality follows from (38).

This establishes (32) for all t. □

Remark 16 (Graceful Degradation)**.** *When $\mathcal{M}_{\text{safe}}(t) = \varnothing$ (no mode satisfies all specifications), the EMERGENCY mode m_5 reuses the most recent valid prediction, trading inference freshness for guaranteed availability. Staleness is bounded to at most $\tau_{\min}$ steps under the dwell-time constraint. The safety guarantee* (32) *still applies to the last step at which a non-emergency mode was selected; during emergency operation, the thermal bound remains valid but the network coverage may be degraded until a safe mode becomes available.*

8 Supplementary Lemmas

Lemma 17 (Monotonic Deque Invariant)**.** *At every step t, Algorithm 1 maintains the invariant that the entries in the deque Q are (i) sorted in non-decreasing order of robustness value from front to back, and (ii) have timestamps within $[t - W, t]$.*

Proof. By induction on t.

Base case ($t = 0$): The deque is initialized with the single entry $(0, \mu(\mathbf{x}_0))$, trivially satisfying both conditions.

Inductive step: Assume the invariant holds at step $t - 1$. At step t:

1. The new value $\mu(\mathbf{x}_t)$ is computed.
2. Entries with robustness $\geq \mu(\mathbf{x}_t)$ are removed from the back (lines 4–5 of Algorithm 1). This preserves the non-decreasing order: after removal, the back entry has robustness $< \mu(\mathbf{x}_t)$, and the new entry $\mu(\mathbf{x}_t)$ appended to the back maintains the ordering.
3. Entries with timestamps $< t - W$ are removed from the front (lines 7–8). Since entries were inserted in chronological order and the deque is a FIFO structure from the front, all remaining entries have timestamps $\geq t - W$.
4. After these operations, all entries have timestamps in $[t - W, t]$ and robustness values in non-decreasing order.

The front element thus holds the minimum robustness over $[t - W, t]$, yielding:

$$Q.\text{front.value} = \min_{k \in [0, W]} \mu(\mathbf{x}_{t-k}) = \rho(\boxminus_{[0,W]} \varphi, \sigma, t). \tag{40}$$

□

Lemma 18 (EMA Convergence)**.** *If the true KL divergence $D_{\text{KL}}(\mathcal{P}_{\text{run}} \| \mathcal{P}_{\text{cal}})$ stabilizes at a constant value D^* for $t \geq t_0$, then the EMA estimate $\hat{D}_t$ converges geometrically:*

$$|\hat{D}_t - D^*| \leq (1 - \alpha)^{t - t_0} \cdot |\hat{D}_{t_0} - D^*|, \tag{41}$$

where $\alpha \in (0, 1)$ is the EMA decay parameter.

Proof. For $t > t_0$, assuming the batch estimate equals the true divergence ($D_{\mathrm{KL}}(\text{batch}) = D^*$):

$$\hat{D}_t - D^* = (1-\alpha)\hat{D}_{t-1} + \alpha D^* - D^* \tag{42}$$

$$= (1-\alpha)(\hat{D}_{t-1} - D^*). \tag{43}$$

Unrolling the recursion:

$$\hat{D}_t - D^* = (1-\alpha)^{t-t_0}(\hat{D}_{t_0} - D^*). \tag{44}$$

Since $\alpha \in (0,1)$, $|(1-\alpha)^{t-t_0}| < 1$ and converges to 0 geometrically. □

References

[1] A. Donzé and O. Maler, "Robust satisfaction of temporal logic over real-valued signals," in *Proc. FORMATS*, 2010.

[2] G. E. Fainekos and G. J. Pappas, "Robustness of temporal logic specifications for continuous-time signals," *Theoretical Computer Science*, vol. 410, no. 42, pp. 4262–4291, 2009.

[3] O. Maler and D. Nicković, "Monitoring temporal properties of continuous signals," in *Proc. FORMATS/FTRTFT*, 2004.

[4] A. B. Tsybakov, *Introduction to Nonparametric Estimation*. Springer, 2009.

[5] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World*. Springer, 2005.

[6] Y. Zhao, B. Hoxha, G. Fainekos, J. V. Deshmukh, and L. Lindemann, "Robust conformal prediction for STL runtime verification under distribution shift," in *Proc. ACM/IEEE ICCPS*, 2024.

[7] M. Sugiyama, T. Suzuki, and T. Kanamori, *Density Ratio Estimation in Machine Learning*. Cambridge University Press, 2012.

[8] C. Rowe et al., "Efficient online monitoring of signal temporal logic specifications via sliding window algorithms," 2023.